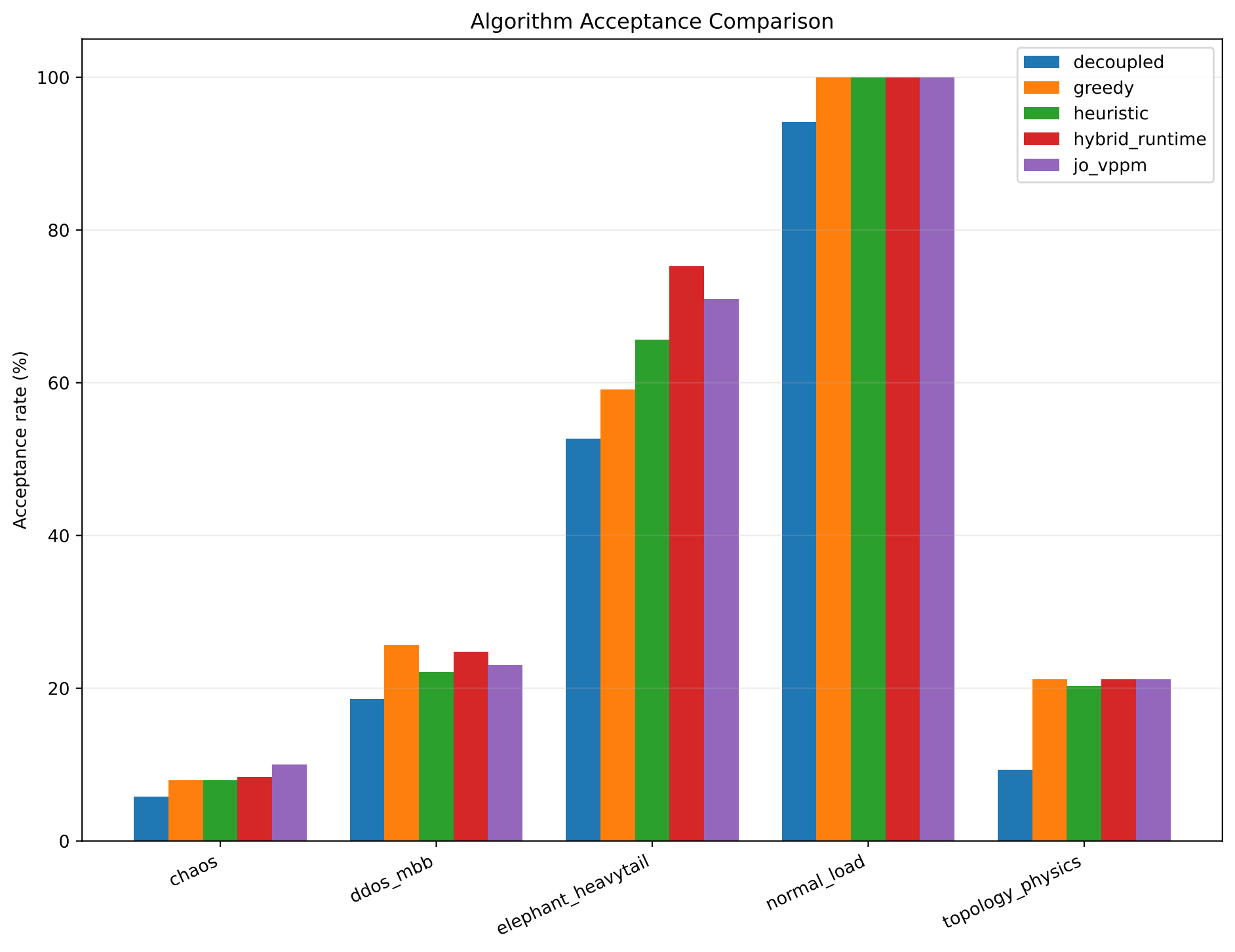
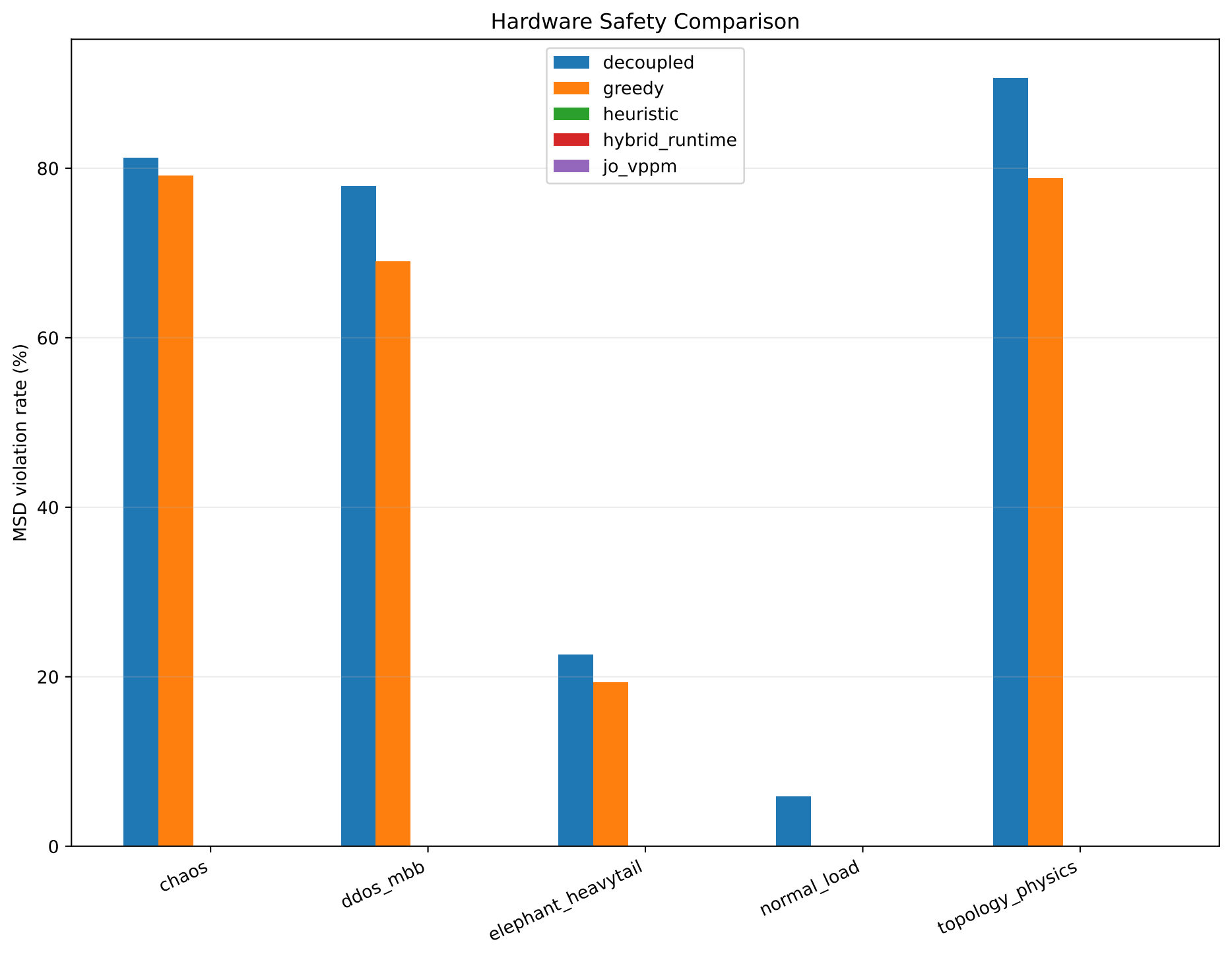
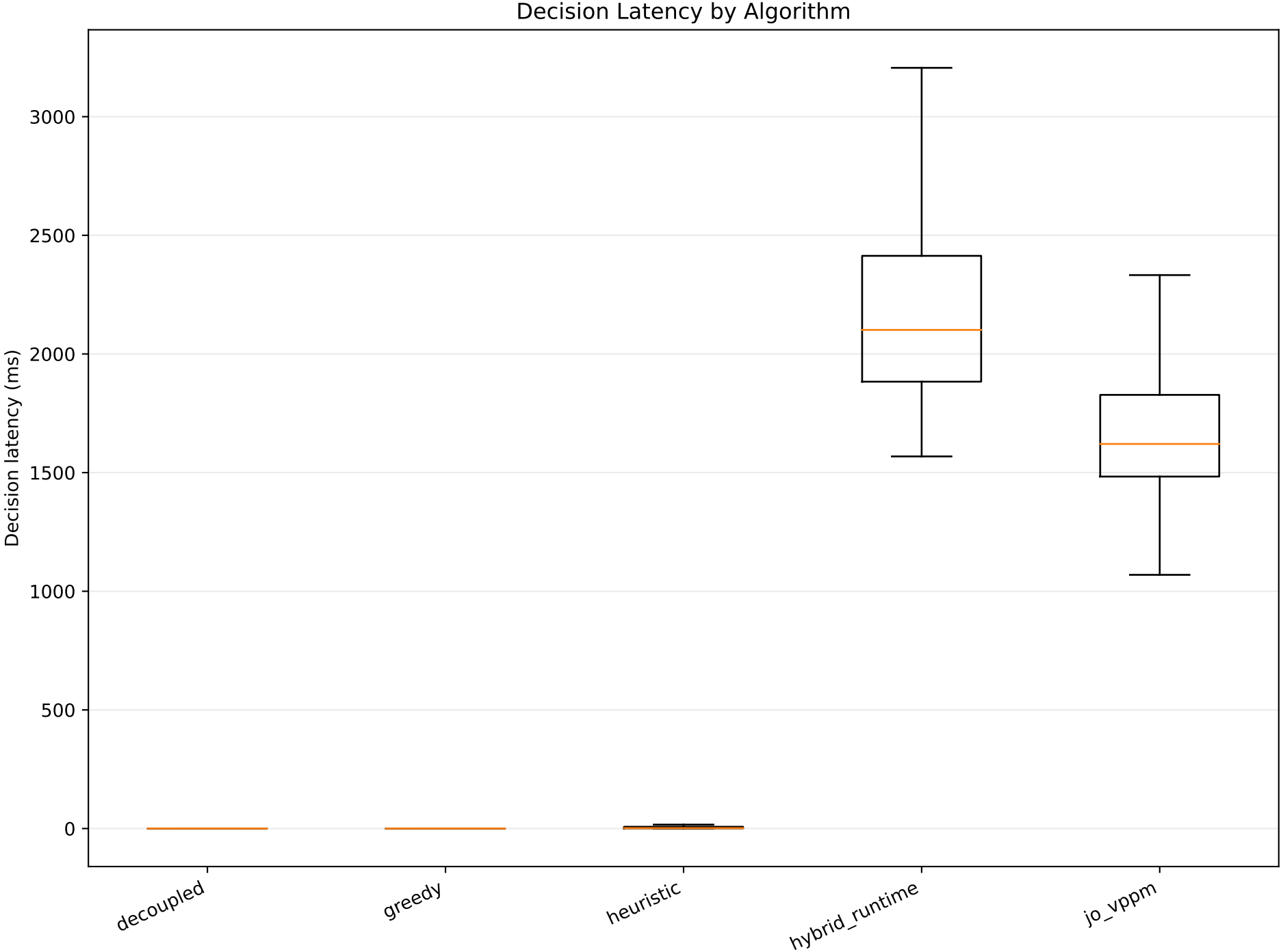
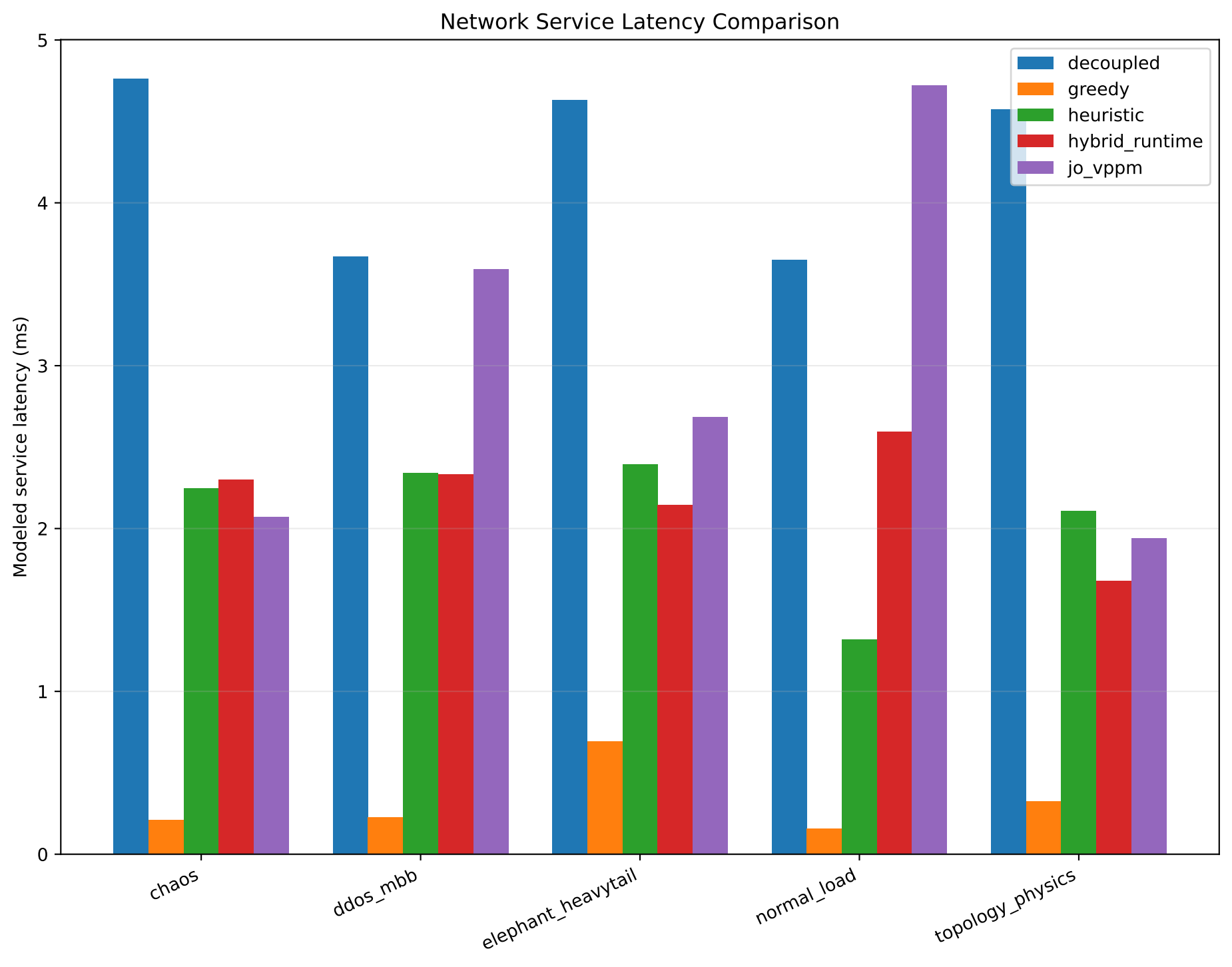
3S-COM Testbed Benchmark Report

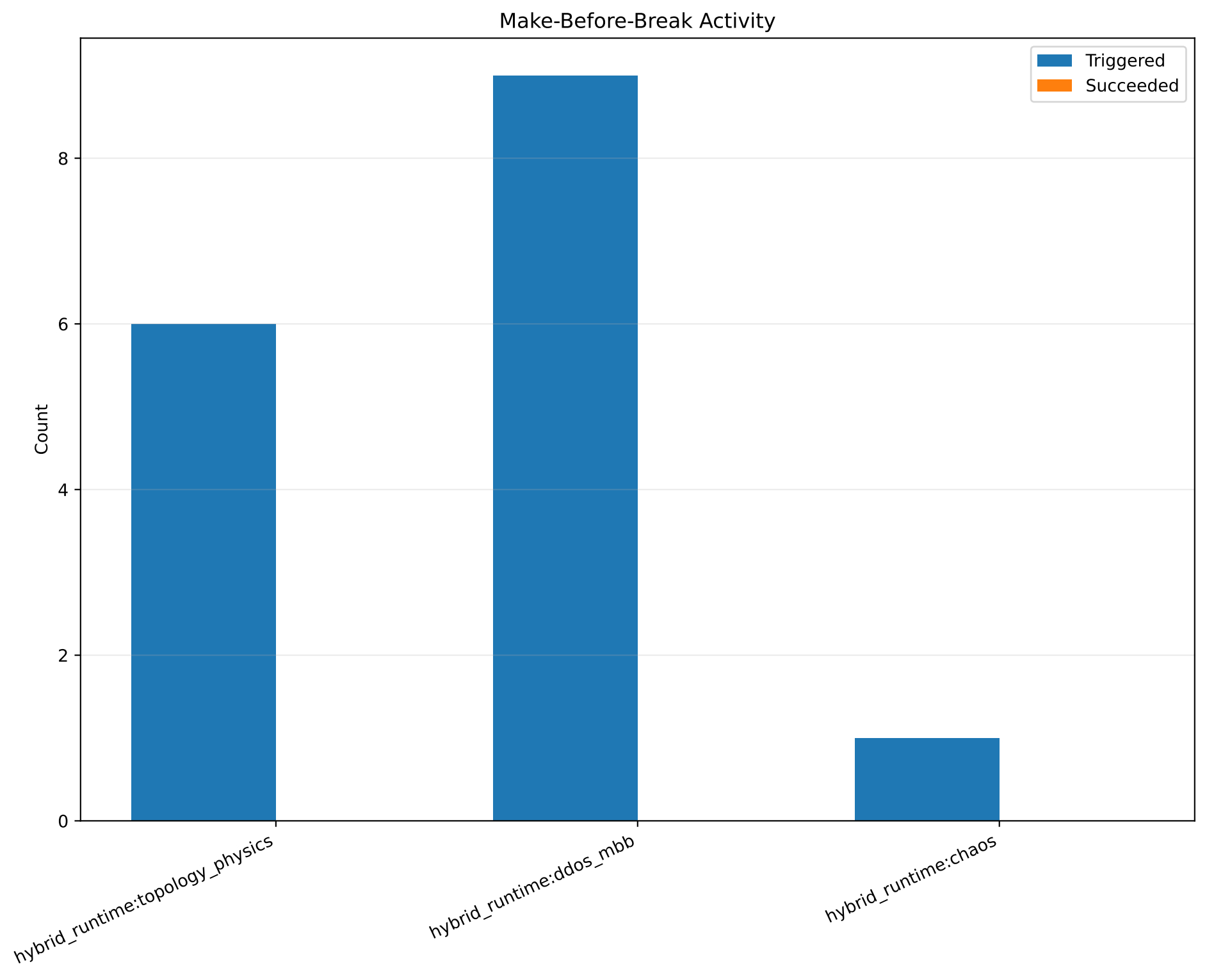
Scenario	Algorithm	Requests	Accept	No safe	MSD viol	Decision ms	Service ms	Timeouts	Migrations	MSD drops
nario 1 - Normal	hybrid_runtime	34	100.0%	0.0%	0.0%	259.2	2.60	0	0	0
nario 1 - Normal	jo_vppm	34	100.0%	0.0%	0.0%	1588.0	4.72	0	0	0
nario 1 - Normal	heuristic	34	100.0%	0.0%	0.0%	27.5	1.32	0	0	0
nario 1 - Normal	greedy	34	100.0%	0.0%	0.0%	0.6	0.16	0	0	0
nario 1 - Normal	decoupled	34	94.1%	0.0%	5.9%	0.3	3.65	0	0	0
io 2 - Elephant H	hybrid_runtime	93	75.3%	24.7%	0.0%	2011.4	2.14	0	0	0
io 2 - Elephant H	jo_vppm	93	71.0%	29.0%	0.0%	1823.9	2.69	0	0	0
io 2 - Elephant H	heuristic	93	65.6%	34.4%	0.0%	5.9	2.40	0	0	0
io 2 - Elephant H	greedy	93	59.1%	21.5%	19.4%	0.6	0.70	0	0	0
io 2 - Elephant H	decoupled	93	52.7%	0.0%	22.6%	0.4	4.63	0	0	0
- Topology and P	hybrid_runtime	118	21.2%	78.8%	0.0%	3992.8	1.68	0	6	0
- Topology and P	jo_vppm	118	21.2%	78.8%	0.0%	1868.6	1.94	0	0	0
- Topology and P	heuristic	118	20.3%	79.7%	0.0%	9.7	2.11	0	0	0
- Topology and P	greedy	118	21.2%	0.0%	78.8%	0.4	0.32	0	0	0
- Topology and P	decoupled	118	9.3%	0.0%	90.7%	0.7	4.57	0	0	0
DDoS and Make	hybrid_runtime	113	24.8%	75.2%	0.0%	4632.4	2.33	0	9	0
DDoS and Make	jo_vppm	113	23.0%	77.0%	0.0%	1671.3	3.59	0	0	0
DDoS and Make	heuristic	113	22.1%	77.9%	0.0%	4.7	2.34	0	0	0
DDoS and Make	greedy	113	25.7%	5.3%	69.0%	3.0	0.23	0	0	0
DDoS and Make	decoupled	113	18.6%	0.0%	77.9%	0.4	3.67	0	0	0
Scenario 5 - Cha	hybrid_runtime	240	8.3%	91.7%	0.0%	2180.9	2.30	0	1	0
Scenario 5 - Cha	jo_vppm	240	10.0%	90.0%	0.0%	1640.9	2.07	0	0	0
Scenario 5 - Cha	heuristic	240	7.9%	92.1%	0.0%	9.0	2.25	0	0	0
Scenario 5 - Cha	greedy	240	7.9%	12.9%	79.2%	1.3	0.21	0	0	0
Scenario 5 - Cha	decoupled	240	5.8%	0.0%	81.2%	1.8	4.76	0	0	0


Algorithm Acceptance Comparison
Acceptance rate (%)
0
20
40
60
80
100
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
chaos
ddos_mbb
elephant_heavytail
normal_load
topology_physics


Hardware Safety Comparison
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
MSD violation rate (%)
0
20
40
60
80
chaos
ddos_mbb
elephant_heavytail
normal_load
topology_physics


Decision Latency by Algorithm
Decision latency (ms)
3000
2500
2000
1500
1000
500
0
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm


Network Service Latency Comparison
Modeled service latency (ms)
0
1
2
3
4
5
decoupled
greedy
heuristic
hybrid_runtime
jo_vppm
chaos
ddos_mbb
elephant_heavytail
normal_load
topology_physics


Make-Before-Break Activity
Triggered
Succeeded
Count
0
2
4
6
8
hybrid_runtime:topology_physics
hybrid_runtime:ddos_mbb
hybrid_runtime:chaos